

Secure Adaptive Continuum Explorer

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TUE-16



VEERESH S K (23B1309)
RISHITH GUPTA (23B1234)
ARSHIT SINGH (23B1275)
PRITHVI SANGAM (23B1292)
ARIJIT PAUL (23B1219)

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SUMMARY

01

WHAT?

modular continuum tentacle-based end-effector
using tendon-driven actuation and visual feedback

02

WHO?

industrial maintenance, medical/surgical researchers
and rescue professionals requiring flexible robotic
systems

03

PROBLEM?

overcomes limited accessibility and dexterity,
adaptive grasping and safe interaction

SPECIFICATIONS

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01

LINK BENDING : 30°

04

MINIMUM GRABBING : 1.25 cm
RADIUS

02

EXTENSION OF : 40 cm
LEADSCREW

05

MAXIMUM GRABBING : 3.5 cm
RADIUS

03

CAMERA FOV : 60°

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STATUS OF PROJECT

Key Requirements proposed at the beginning of the semester	Status at end of semester	Remarks/Justification
bending of individual continuum links	complete	
end-effector object gripping	complete	
hybrid continuum-tentacle mechanical design	complete	

STATUS OF PROJECT



Key Requirements proposed at the beginning of the semester	Status at end of semester	Remarks/Justification
user control through GUI commands on keyboard with live camera feedback	complete	
limit switch-based obstacle avoidance	complete	

STATUS OF PROJECT



Key Requirements proposed at the beginning of the semester	Status at end of semester	Remarks/Justification
integrated multiple RPis and ICs	complete	
autonomous navigation, control and object retireval	incomplete	manual control algorithm complex, and more costly sensors required

PHOTOGRAPHS OF FINAL PROTOTYPE

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PILLOW
BEARING

LEADSCREW

× × × ×

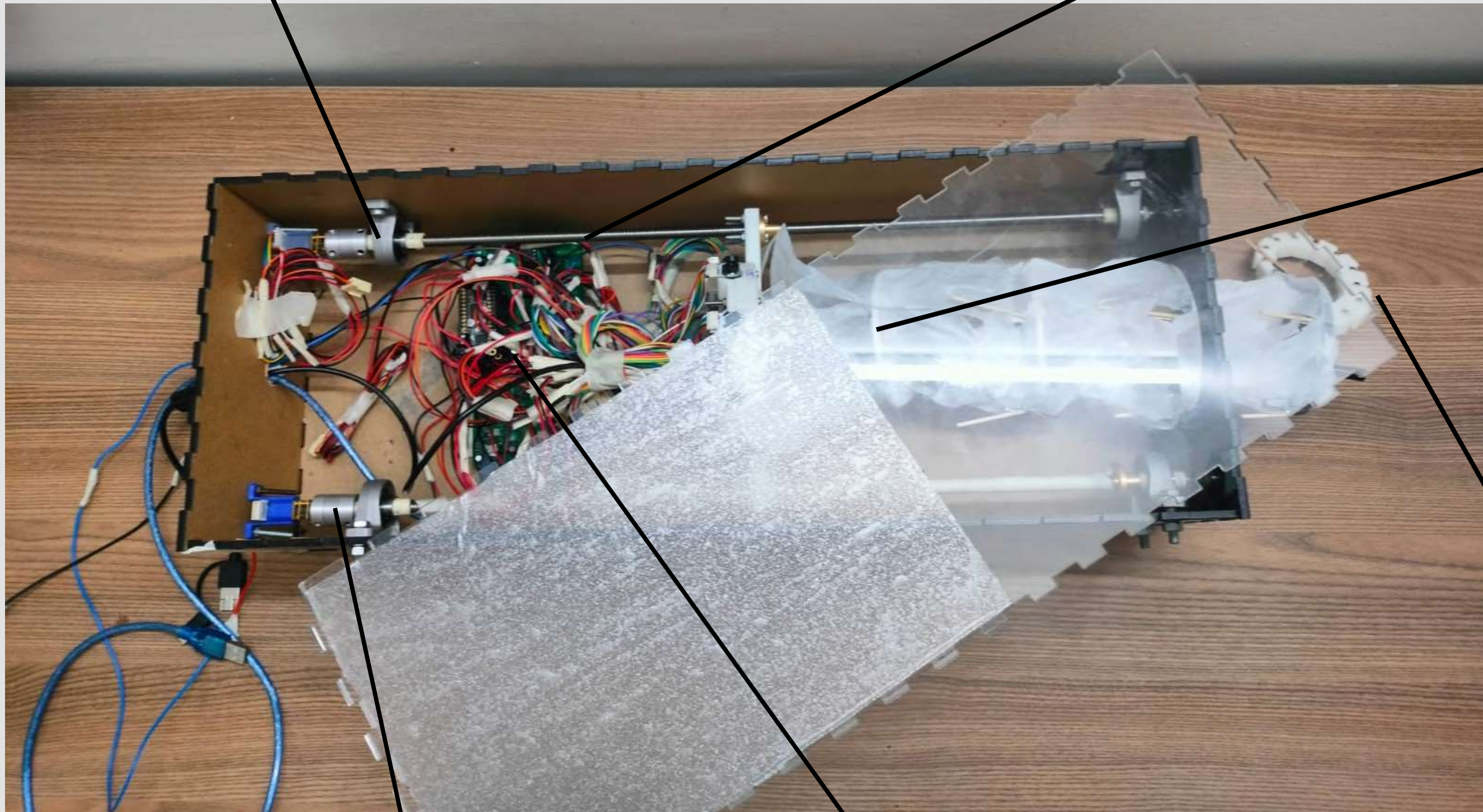
CONTINUUM
LINKS

TENTA
END-EF E
ELE
CTOR

COUPLERS

PCB

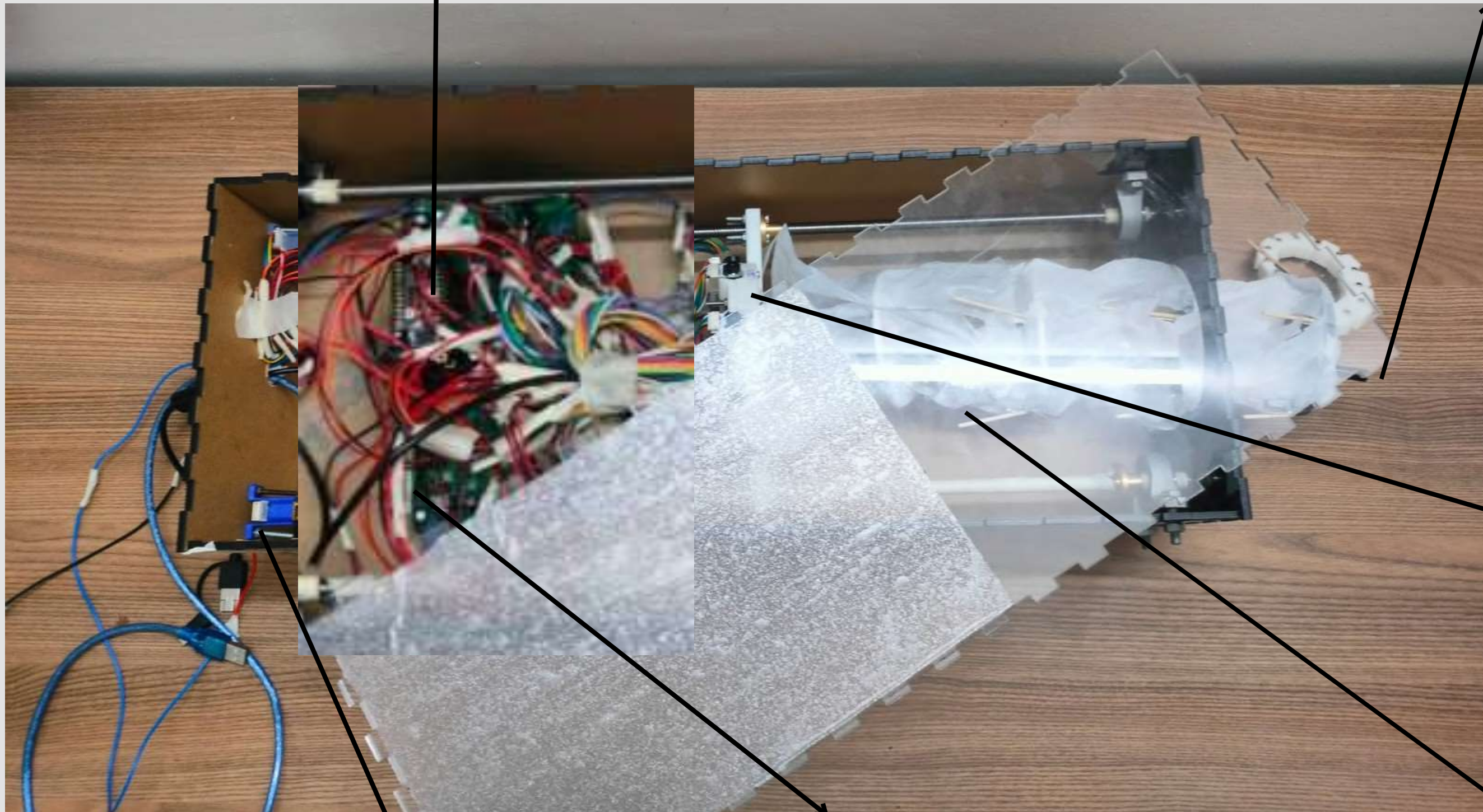
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MOTOR DRIVER

ENDOSCOPE
CAMERA

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BASE PLATE
WITH MOTORS

LIMIT SWITCHES

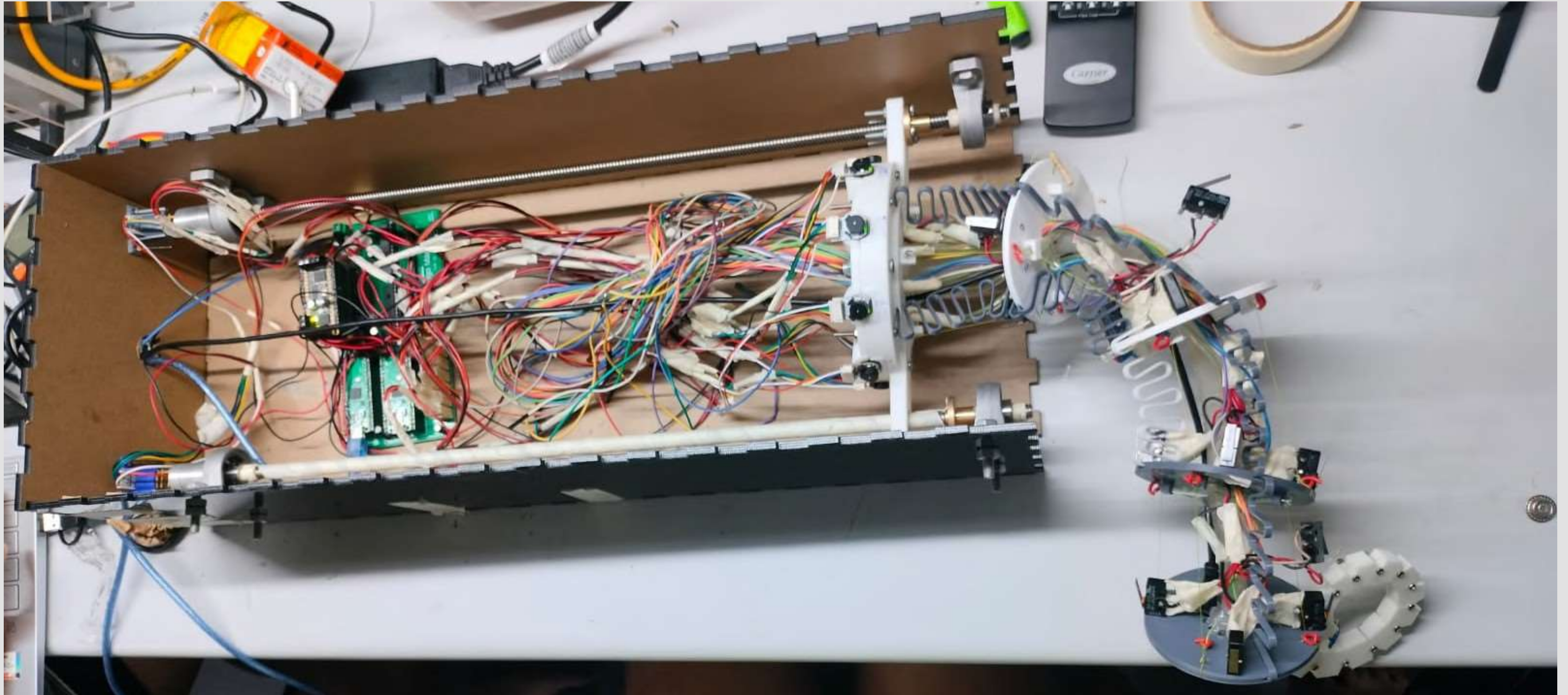
LEADSCREW
MOTOR

MASTER RPI

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CONTINUUM BENDING TO THE LIMIT !!!

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Challenges	Solutions	Remarks	CHALLENGES FACED AND HOW WE SOLVED THEM
spool design	multiple iterations, 2 winding spool with opposite directions	working but better solutions possible	
tentacle material choice and design	final design hinge-based, printed in PLA, tried resin printing	effective bending and movement	
insufficient ports on RPI Pico	used port expanders and master-slave architecture	increased ports (26 to 26+32) and better task distribution	
realised torque was less	changed routing of cables and brought motors to base plate	getting enough torque	
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Challenges	Solutions	Remarks	CHALLENGES FACED AND HOW WE SOLVED THEM
mistakes in PCB design	fixed using micro-surgery and removal of Vsys of main RPi	PCB is working without any issues	
defect in motors	micro-surgery to connect terminals and connector	motor defect fixed and working	
double lead screw alignment	keeping one lead screw stationary, moving other	better movement than earlier	
tight mechanical tolerances	changed designs and 3D printers for better prototype	more flexible and accurate design	

DEMO VIDEO



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**MINDS
BEHIND
THIS**

